

ENGINEERING GRAPHICS

Unit – 1 – PLANE CURVES AND PROJECTION OF POINTS

Syllabus: Curves used in engineering practices: Conics – Construction of ellipse, parabola and hyperbola by eccentricity method – Cycloidal Curves – Construction of cycloid, epicycloid and hypocycloid – Construction of involutes of square and circle – Drawing of tangents and normal to the above curves. Principles of Projection and Projection of points.

1. Introduction to Conics – 01 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/LiHYpiTIVBQ>

2. Draw a parabola when the distance between focus and directrix is 60 mm. Draw a tangent and normal at any point on the curve.

Parabola – 02 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/XEUWz-2Y7Xk>

3. Draw a parabola when the distance between focus and directrix is 50 mm. Draw a tangent and normal at a point distance 70 mm from the directrix.

Parabola – 03 | Engineering Graphics Made Easy (CAD Method)

Link: https://youtu.be/_qB_COYb9qg

4. Draw the curve, when the distance between focus and vertex is 50 mm and the eccentricity $e=3/2$ Draw a tangent and normal at any point on the curve.

Hyperbola – 04 | Engineering Graphics Made Easy (CAD Method)

Link: https://youtu.be/Y_Gfkyt5xfA

5. The focus of a hyperbola is 60 mm from its directrix. Draw the curve when eccentricity is $5/3$. Draw a tangent and a normal to the curve at a point 45 mm from the directrix.

Hyperbola – 05 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/JZrZcWTCY9A>

6. Construct an ellipse with a distance of the focus from the directrix which is 70 mm and the eccentricity which is $2/3$. Also draw a normal and tangent to a point on the ellipse which is at a distance of 70 mm from the directrix.

Ellipse – 06 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/-GEB3Hb2SZE>

7. A Circle of 50mm diameter rolls on a straight line without slipping. Trace the locus of a point "P" on the circumference of the circle rolling for one revolution. Name the curve. Draw normal and tangent to the curve at any point on the curve.

Cycloid – 07 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/ISwm61ldkI8>

8. A coin of 40mm diameter rolls over a horizontal table without slipping. A point on the circumference of the coin is in contact within the table surface. In the beginning and after one complete revolution. Draw the path traced by the coin. Draw a tangent and normal at any point on the curve.
9. Draw an epicycloid of a circle having a diameter of 50 mm, which rolls outside another circle of 150 mm diameter for one complete revolution. Also, draw a tangent and normal to the epicycloid at a point 120 mm from the centre of a directing circle.

Epicycloid – 09 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/MPmWdSWbqAA>

10. Construct an epicycloid of a rolling circle 60 mm diameter which rolls outside of directing circle 122 mm radius for one revolution. Draw a tangent and normal to any point on the curve.

Epicycloid – 10 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/1NRuD7vdRXs>

11. Draw locus of a point on the periphery of a circle which rolls from the inside of a curved path. Take diameter of rolling circle 50 mm and radius of directing circle (curved path) 75 mm. Also, draw a tangent and normal at any point on the curve.

Hypocycloid – 11 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/AQgCbdfkVtE>

12. Draw an involute of a square with a sides of 20 mm. Also draw a tangent and normal at any point on the curve.

Involute of Square – 12 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/GJq8KwyDY-Y>

13. Draw an Involute of a circle of 50 mm Diameter. String length is equal to the circumference of a circle. Also, draw a normal and tangent at anywhere on the curve.

Involute of Circle – 13 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/LQoIgiTpQJM>

14. Draw the orthographic projections of the following points, take 20 mm distance between successive projections in a single reference line:

- a) Point A is 25 mm above H.P and 30 mm in front of VP
- b) Point B is 20 mm above H.P and 35 mm behind VP
- c) Point C is 32 mm below H.P and 25 mm behind VP
- d) Point D is 35 mm below H.P and 20 mm in front of VP
- e) Point E is on H.P and 25 mm behind VP
- f) Point F is in V.P and 40 mm below HP
- g) Point G is in V.P and 37 mm above H.P
- h) Point H is on H.P and 35 mm in front of VP
- i) Point I on H.P and in VP.

Projection of Points – 14 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/TRtfYkiW0Z4>

15. The point A is in HP and 25 mm in front of VP another point B is also in HP and Behind VP. The distance between end projectors is 55 mm. Draw its projection when the line joining the plan makes an angle of 50 degree also find the distance of the point from VP.

Projection of Points – 15 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/PQ5J7ujBq1Q>

Unit – 2 – PROJECTION OF LINES AND PLANE SURFACES

Syllabus: Projection of straight lines (First angle projection) inclined to both the principal planes – Determination of true lengths and true inclinations by rotating line method.

Projection of planes (polygonal and circular surfaces) inclined to both the principal planes by rotating object method.

Projection of Straight Lines

1. A line AB 70mm line whose one end is 20mm above HP and 30mm In front of VP. The Line is inclined at 30° to HP and 40° to VP. Draw the projection of the line AB.

Projection of Straight Lines – 1 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/Xv0BEny3Qsc>

2. A line BC is 60mm long whose one end be 20mm above HP and 30mm In front of VP. The other end C is 60mm In front of VP and 40mm above HP. Draw the projection line B and C.

Projection of Straight Lines – 2 | Engineering Graphics Made Easy (CAD Method)

Link: https://youtu.be/9GG-ZuRf4_U

3. The line CD whose one end is 20mm above HP and 30mm Infront of VP. The other end will be 60mm above HP and 55mm Infront of VP. The distance between the projectors is 30mm. Draw the projections and find the true length and true inclinations.

Projection of Straight Lines – 3 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/Y54LXyeBj7c>

4. A line EF is 65mm long whose one end is 15mm above HP and 25mm Infront of VP. The top view measures 40mm and makes an angle 35° with VP. Draw the projection of a line EF.

Projection of Straight Lines – 4 | Engineering Graphics Made Easy (CAD Method) - Updated

Link: <https://youtu.be/n5p1LOqQuyw>

5. A line DE is 60mm long whose front view measures 50mm and top view measures 40mm. One end of the line D is 15mm above HP and 25mm Infront of VP. Draw the projections and find the true inclination of line DE with HP and VP.

Projection of Straight Lines – 5 | Engineering Graphics Made Easy (CAD Method) - Updated

Link: <https://youtu.be/hxj66Dkd1Ok>

Important Previous Year End Semester Questions for Self-Practice:

6. One end of the line CD 70 mm long is 20 mm above HP and 15 mm in-front of VP. It is inclined at 40° to VP and 50° to HP. Draw its projections.

Projection of Straight Lines – 6 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/3wZvwDhkRuU>

7. A 70 mm long line LM has its end L, 10 mm above HP and 15 mm in front of VP. The top view and front views of the line measure 60 mm and 40 mm respectively. Draw the projections of the line and determine its inclinations with both HP and VP.

Projection of Straight Lines – 7 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/PIW94jmidzQ>

Projection of Planes

1. A square lamina PQRS of side 40 mm rests on the ground on its corner P in such a way that the diagonal PR is inclined at 45° to the HP and apparently inclined at 30° to the VP. Draw its projections.

Projection of Planes – 1 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/pExhAdy7EI0>

2. A pentagonal plate 30 mm is resting on HP on one its corners with its surface inclined at 45° to HP. The side opposite to the resting corner is parallel to VP and farther away from it. Draw its projections.

Projection of Planes – 2 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/v-IWryUMRNA>

3. A hexagonal plate of 25 mm side is resting on HP such that one of its corners touches both HP and VP. It makes 30° with HP and 60° with VP. Draw the projections by change of position method.

Projection of Planes – 3 | Engineering Graphics Made Easy (CAD Method)

Link: https://youtu.be/UI_vYEZLttQ

4. A circular plate of diameter 50 mm is resting on the HP on a point on the circumference. Its surface is kept inclined at 45° to HP. Draw its projections when the line representing its diameter and passing through the resting point is inclined at 30° to the VP.

Projection of Planes – 4 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/uvm3SXvwm40>

5. A rectangle plate 70 * 40 mm has one of its shorter edge in the VP inclined at 40° to the HP. Draw its projections if its front view is a square of side 40 mm.

Projection of Planes – 5 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/vdKbTmsywGM>

Important Previous Year End Semester Questions for Self-Practice:

6. A hexagonal plate of side 20 mm rests on the HP on one of its sides inclined at 45° to VP. The surface of the plate makes an angle of 30° with the HP. Draw the front and top views of the plate.

Projection of Planes – 6 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/lXjqL1fcfF8>

Unit – 3 – PROJECTION OF SOLIDS AND PROJECTION OF SECTIONED SOLIDS

Syllabus: Projection of simple solids like prisms, pyramids, cylinder and cone when the axis is inclined to one of the principal planes by rotating object method.

Sectioning of solids in simple vertical position when the cutting plane is inclined to HP and perpendicular to VP – obtaining true shape of the section. Practicing 3D modeling of simple objects by CAD software (Not for examination)

Projection of Solids

1. A right regular pentagonal prism, side of base 30 mm and height of axis as 75 mm rests on HP on one of its base corners such that its long edge containing the corner is inclined to the HP at 60° . Draw its projections.

Projection of Solids – 1 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/Ahgk9SULnZo>

2. A regular pentagonal prism of 25mm long edges and axis 70mm long rests on HP on one of its corners of the base. The slant edge passing through corner makes 45° . Draw its projections.

Projection of Solids – 2 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/3HlbXw6xv20>

3. A pentagonal prism side of base 25 mm and axis 50 mm rest with one of its edges on HP. Such that the base containing the edge makes an angle 30° to HP and its axis is parallel to VP. Draw its Projections.

Projection of Solids – 3 | Engineering Graphics Made Easy (CAD Method)

Link: https://youtu.be/Vs_DR9j26tw

4. A regular hexagonal prism of 30 mm sides and axis 80 mm long is resting on HP on one of its corners of the base. The axis of the prism makes 30° inclination with HP. Draw its projections.

Projection of Solids – 4 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/RDXS2LL7dYw>

5. A hexagonal prism of base side 30 mm and axis length 60 mm is resting on HP on one of its base corners with a solid diagonal through that corner, which is perpendicular to HP. Draw its projections and print the length of the diagonal.

Projection of Solids – 5 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/9tOtojlod7E>

6. A hexagonal prism of side of base 25 mm and axis 60 mm long lies with one of its rectangular faces on the H.P., such that the axis is inclined at 45° to the V.P. Draw its projections.

Projection of Solids – 6 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/oPkXq87YTsk>

7. A cube of edge 35 mm is resting on H.P on one of its corners with a solid diagonal perpendicular to V.P. Draw the projections of a cube.

Projection of Solids – 7 | Engineering Graphics Made Easy (CAD Method)

Link: https://youtu.be/9ex34NB0_S4

8. Draw the projections of a pentagonal pyramid with a side of base 30 mm and axis 70 mm long when (i) One of its triangular faces is perpendicular to HP and (ii) One of its slant edges is vertical, and (iii) Axis of the pyramid inclined at 60° with HP.

Projection of Solids – 8 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/2oBMq-OhIJ8>

9. A Pentagonal pyramid side of base 20 mm and axis 45 mm long rests with one of its corners on HP such that the base is inclined at an angle of 60° to HP and one side of base is perpendicular to VP. Draw its projections.

Projection of Solids – 9 | Engineering Graphics Made Easy (CAD Method)

Link: https://youtu.be/q_HJNJSKtcc

10. A hexagonal Pyramid of base side 30 mm and axis length 70 mm resting on one of its base edges on HP with axis inclined 30° to HP & parallel to VP. Draw its projections.

Projection of Solids – 10 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/LI4ym-QDYZA>

11. A cone, diameter of base 60 mm and height 70 mm, is resting on HP on the point of periphery of the base. Axis of the cone makes 60° with HP. Draw the projections of the cone, when the apex is nearer to the VP.

Projection of Solids – 11 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/p5d8YKN2iSU>

Projection of Sectioned Solids

1. A pentagonal pyramid of base side 25 mm and altitude 70 mm rests with its base on the HP with a side of base parallel and nearer to the VP. It is cut by a plane inclined at 35° to the HP bisecting the axis. Draw the sectional views.

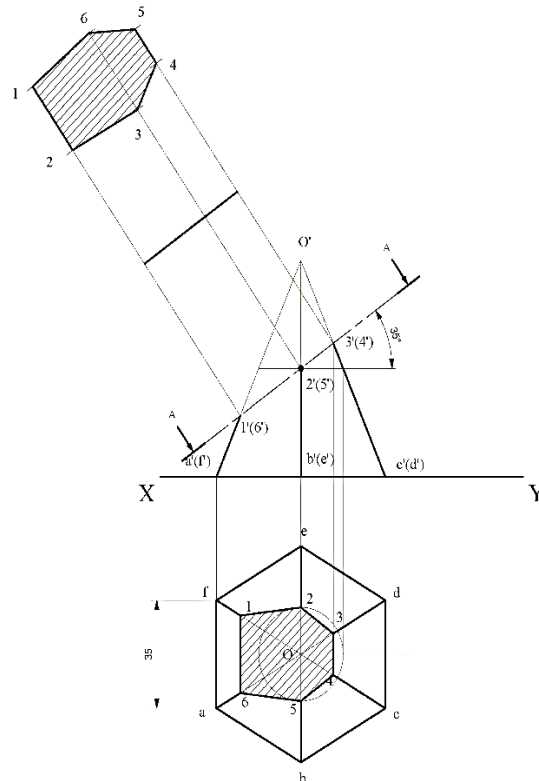
Projection of Sectioned Solids – 1 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/vtc7PjfgMkE>

2. A hexagonal pyramid with a base of 35mm and an altitude of 70mm rests with its base on HP and with a side of the base perpendicular to VP. It is cut by a plane inclined at 35° to HP and bisecting the axis. Draw the sectional plan of the pyramid and the true shape of the section.

Projection of Sectioned Solids – 2 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/Gx4dn7XEPaw>



3. A hexagonal prism of base side 30 mm and axis length 60 mm is resting on HP on one of its bases with two of the vertical faces perpendicular to VP. It is cut by a plane inclined at 60° to HP and perpendicular to VP and passing through a point at a distance 12 mm from the top base. Draw its front view, sectional top view and true shape of section.

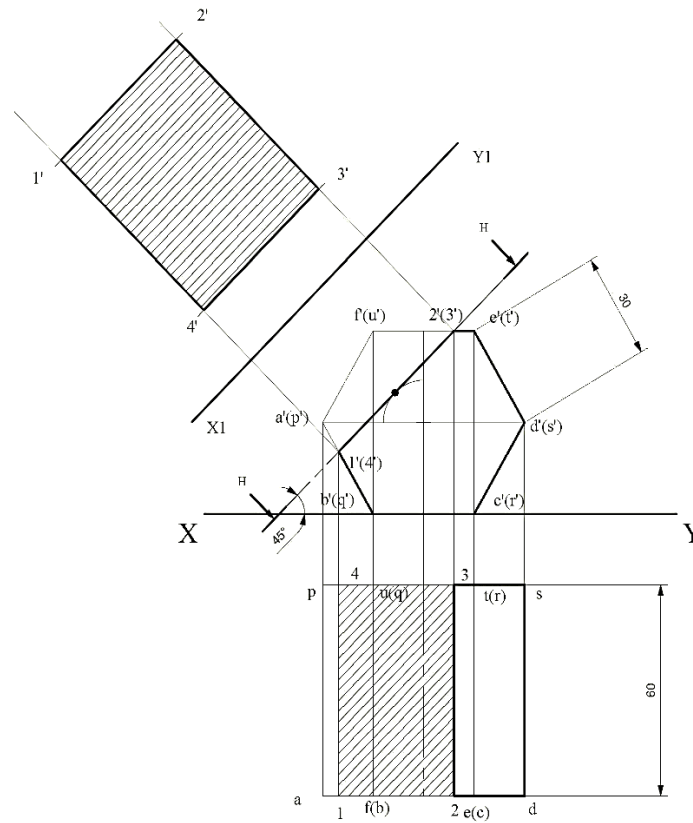
Projection of Sectioned Solids – 3 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/EbJO2D6rc6I>

4. A hexagonal prism of base side 30 mm and axis length 60 mm is resting on HP on one of its rectangular faces with its axis perpendicular to VP. It is cut by a plane inclined at 45° to HP and perpendicular to VP and is 12 mm away from the axis. Draw its front view, sectional top view, and true shape of section.

Projection of Sectioned Solids – 4 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/5sSxj6T8j0E>



5. A cylinder of base diameter 50 mm and height 60 mm rests on its base on HP. It is cut by a plane perpendicular to VP and inclined at 45° to HP. The cutting plane meets the axis at a distance of 15 mm from the top to the base. Draw the sectional plan and true shape of the section.

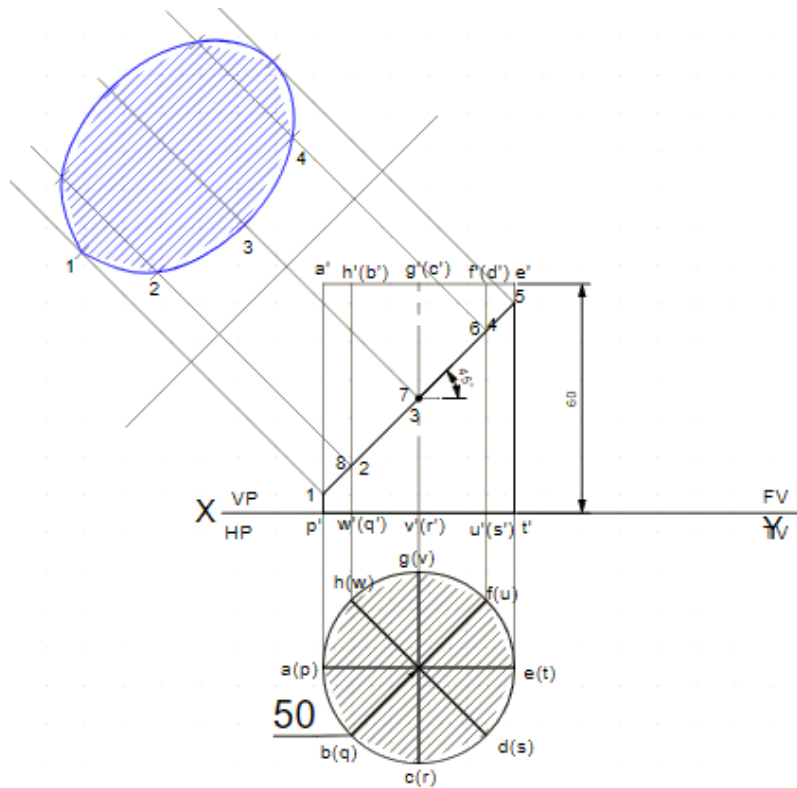
Projection of Sectioned Solids – 5 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/pqbM15qcGFo>

6. A cylinder of diameter 50mm and height of 60mm rests on its base on HP. It is cut by a plane perpendicular to VP and inclined at 45° to HP and bisecting the axis. Draw the sectional front view, top view and true shape of the section.

Projection of Sectioned Solids – 6 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/vz06BDYMjyk>



7. A cone of base 60 mm diameter and axis 70 mm stands vertically with its base on HP. The vertical trace of a section plane is perpendicular to VP and parallel to one of its generators of the cone, passes at a distance of 15 mm from it. Draw its front view, sectional top view and true shape of the section. Name the curve of the true shape of the section.

Projection of Sectioned Solids – 7 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/-xaptbYun4>

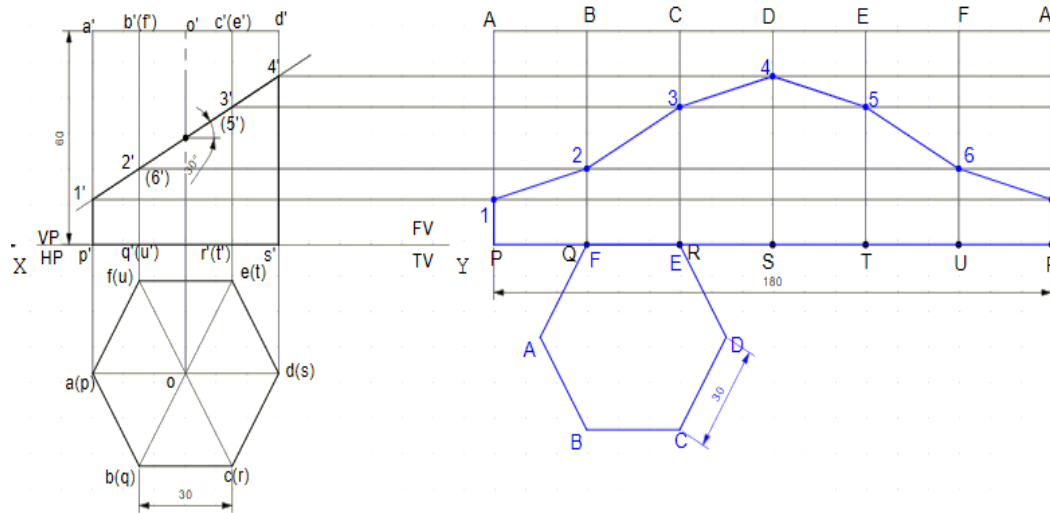
Unit – 4 – DEVELOPMENT OF SURFACES AND ISOMETRIC PROJECTION OF SOLIDS

DEVELOPMENT OF SURFACES

1. A hexagonal prism of base side 30 mm and height 60 mm is resting on the HP with two of its lateral faces parallel to the VP. It is cut by a plane perpendicular to the VP and inclined at 30° to the HP and bisects the axis. Draw the development of the lateral surfaces of the lower portion of the prism.

Development of Surfaces – 1 | Engineering Graphics Made Easy (CAD Method)

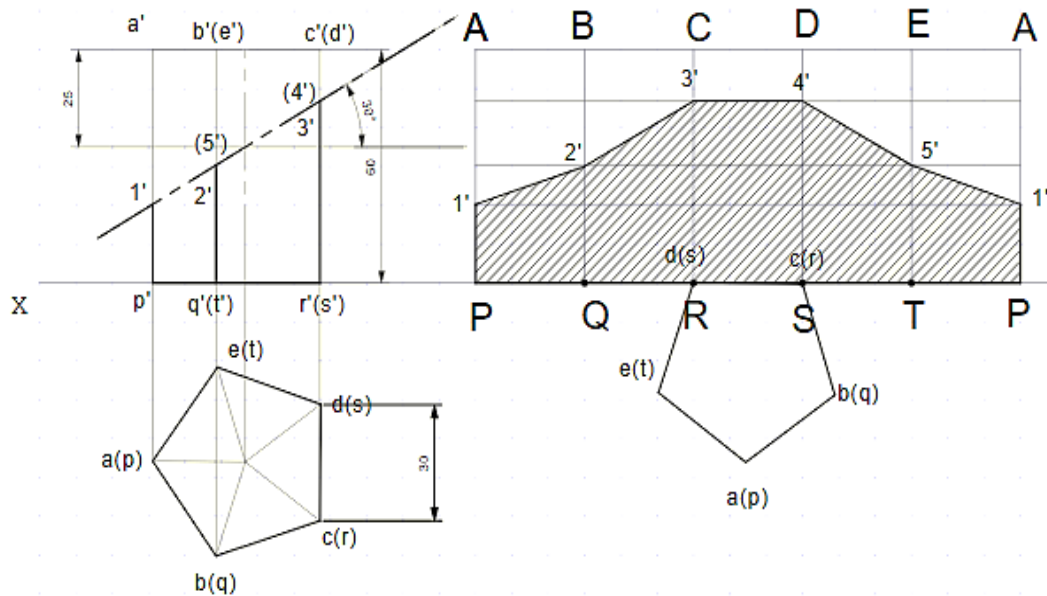
Link: <https://youtu.be/OPEjIhI70z0>



2. A pentagonal prism of base 30 mm and height 60 mm is resting on HP with a base edge perpendicular to VP. A cutting plane perpendicular to V.P and inclined at 30° to HP and 25mm from the top of the prism. Develop the lower portion of the lateral surface of the prism.

Development of Surfaces – 2 | Engineering Graphics Made Easy (CAD Method)

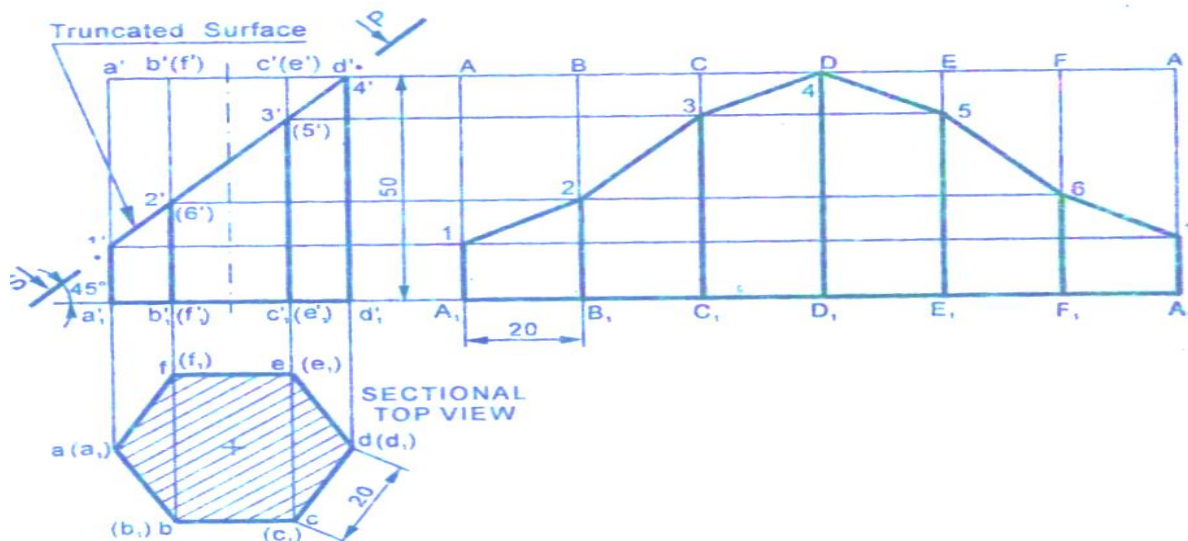
Link: <https://youtu.be/yhPtsonZsjM>



3. A hexagonal prism, edge of base 20 mm and axis 50 mm long, rests with its base on HP such that one of its rectangular faces is parallel to VP. It is cut by a plane perpendicular to VP, inclined at 45° to HP and passing through the right corner of the top face of the prism. Draw the sectional top view and develop the lateral surfaces of the truncated prism.

Development of Surfaces – 3 | Engineering Graphics Made Easy (CAD Method)

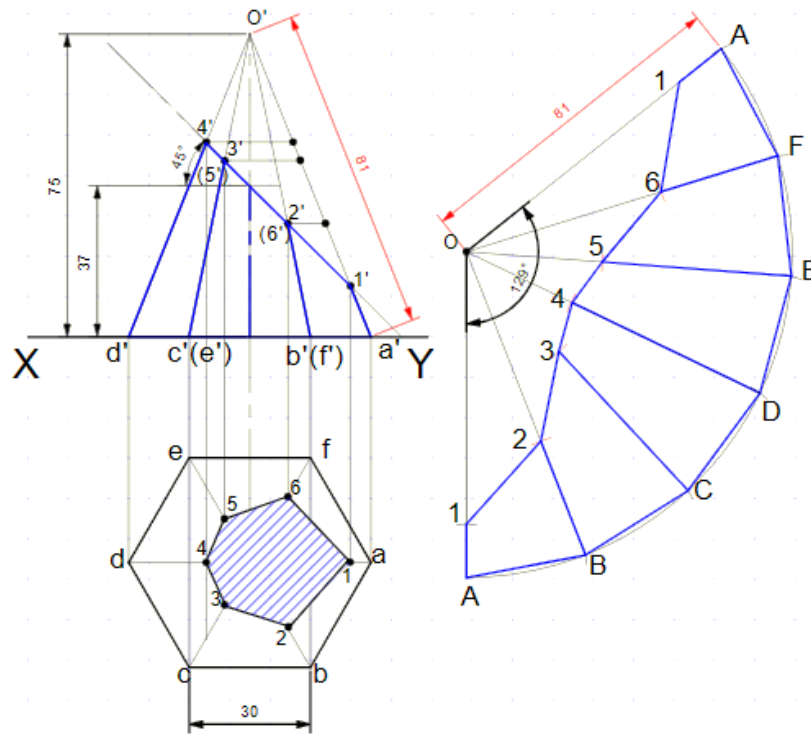
Link: <https://youtu.be/hvPJWlbrumQ>



4. A hexagonal pyramid with a base side of 30 mm and height 75 mm stands with its base on HP and an edge of the base parallel to VP. It is cut by a plane perpendicular to VP, inclined at 45° to H.P and passing through the mid-point of the axis. Draw the (sectioned) top view and develop the lateral surface of the truncated pyramid.

Development of Surfaces – 4 | Engineering Graphics Made Easy (CAD Method)

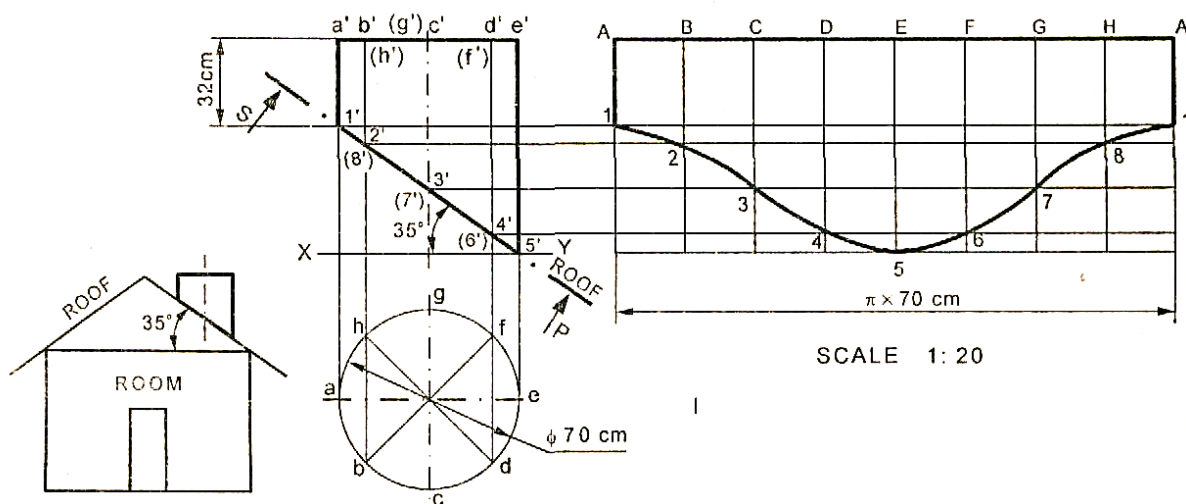
Link: <https://youtu.be/jhaNiW9X1nw>



5. A vertical chimney of 70 cm diameter joins a roof sloping at 35° with horizontal. The shortest portion over the roof is 32 cm. Obtain the shape of the sheet metal from which chimney can be fabricated.

Development of Surfaces – 5 | Engineering Graphics Made Easy (CAD Method)

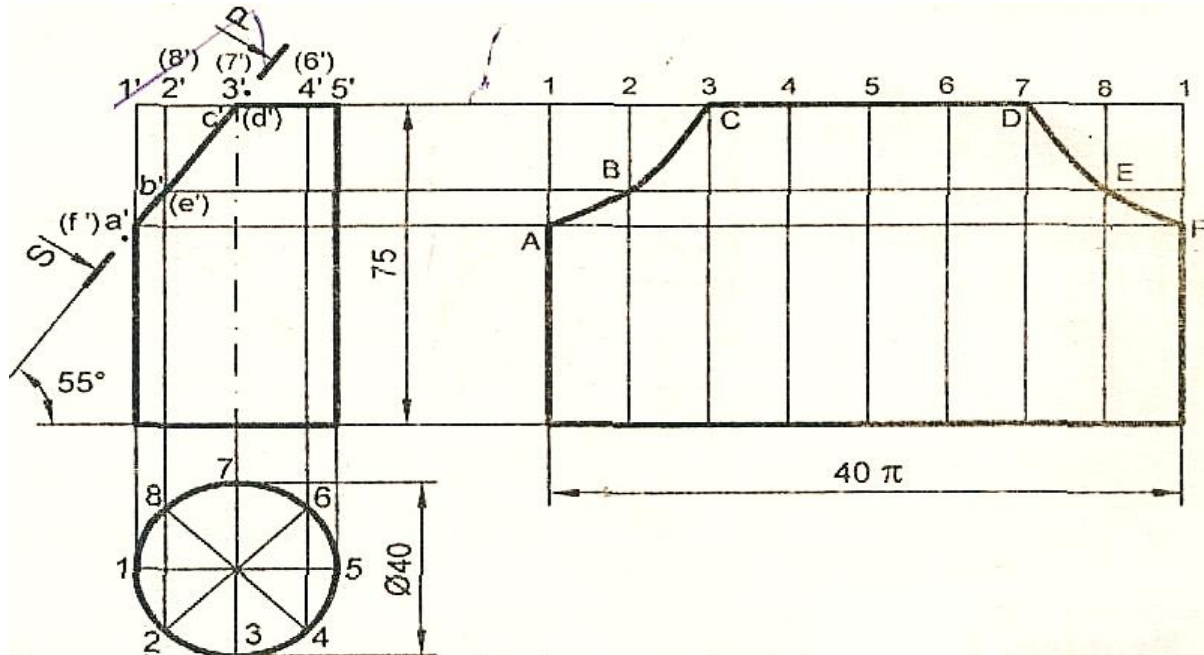
Link: <https://youtu.be/OdXJYMflwms>



6. A plane inclined at 55° to HP cuts a cylinder of diameter 40 mm, height 75 mm and perpendicular to VP is meeting the axis at top face. Draw the lateral development of the solid.

Development of Surfaces – 6 | Engineering Graphics Made Easy (CAD Method)

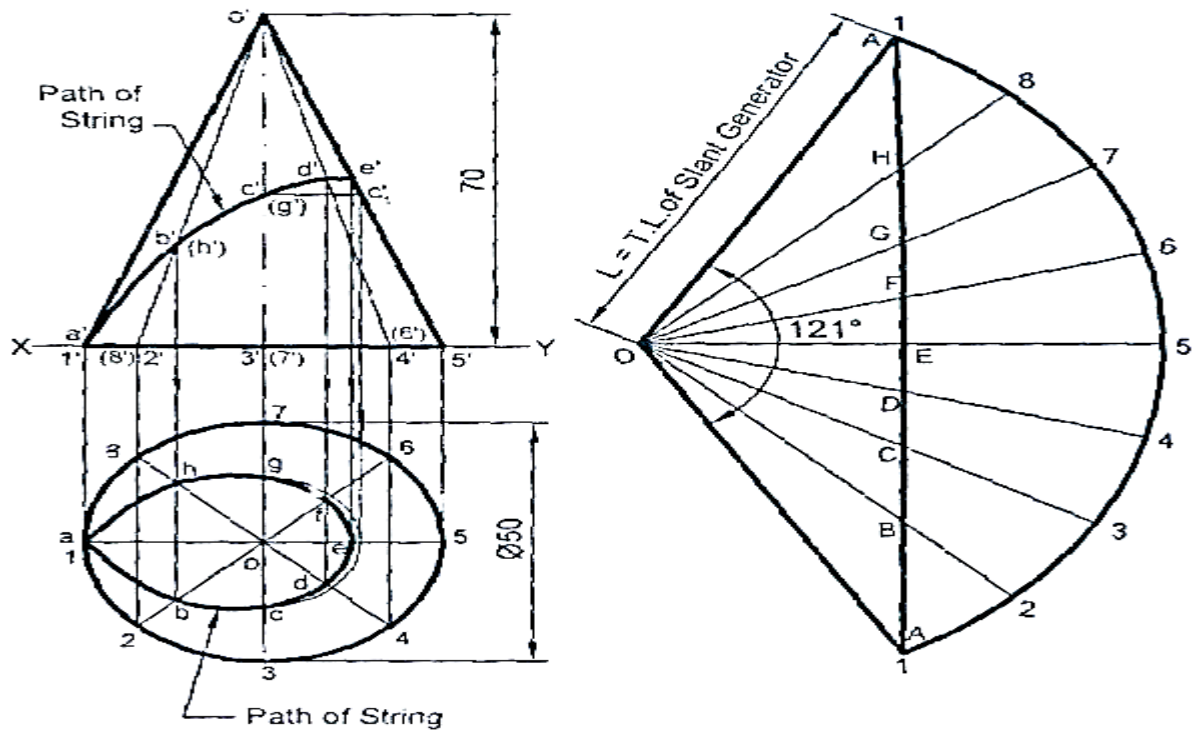
Link: <https://youtu.be/dyFpNVnbCT0>



7. A cone of base diameter 50 mm and height 70 mm rests on its base on ground. A string is wound round the curved surface of the cone starting from left extreme point and ending at the same point. Find the shortest length of the string required. Trace the path of string in front and top views.

Development of Surfaces – 7 | Engineering Graphics Made Easy (CAD Method)

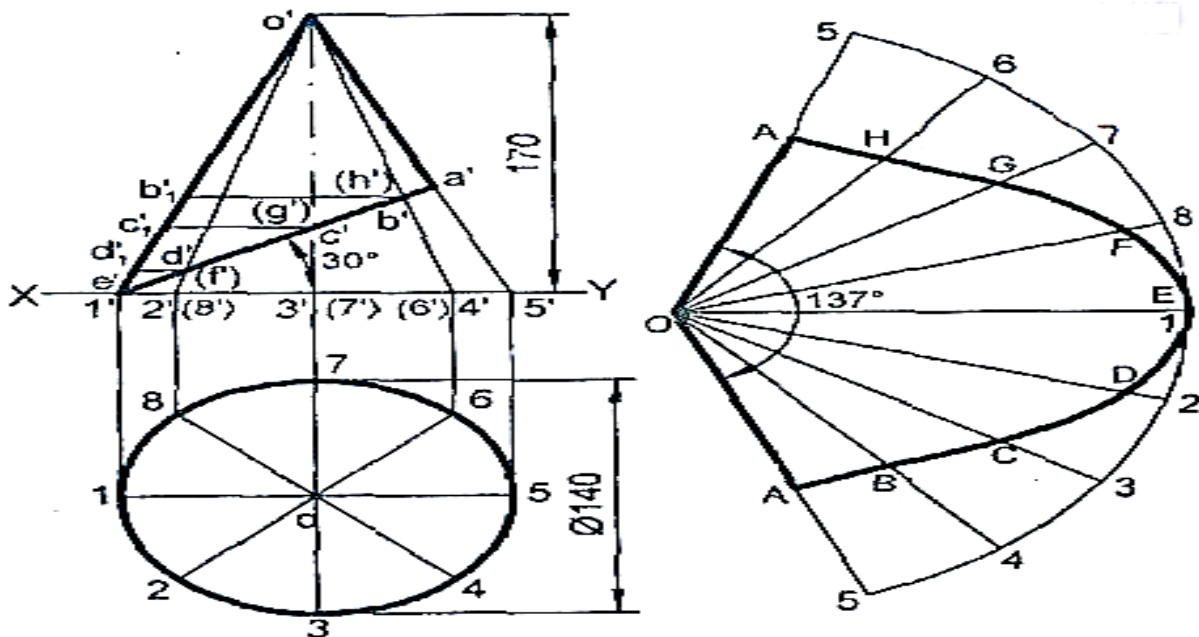
Link: <https://youtu.be/AiKi3hRaS2A>



8. A lampshade is formed by cutting a cone of base 140mm diameter and 170 mm height by a horizontal plane inclined at 30° to HP is passing through one extremity of the base. Develop its lateral surface of the cone.

Development of Surfaces – 8 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/1H2MGJJ1yAs>

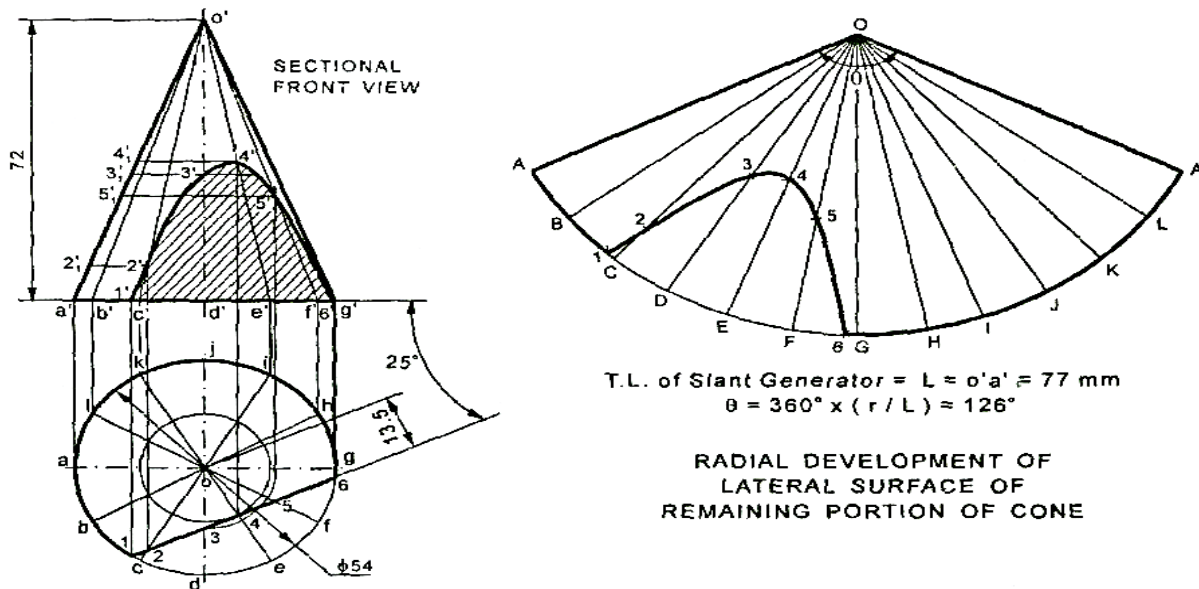


9. A cone, base 54 mm diameter and height 72 mm, rests with its base on HP. A section plane perpendicular to HP and inclined at 25° to VP cuts the cone at a distance of 13.5

mm from the axis. Draw the sectional front view and develop the lateral surface of the remaining portion of the cone.

Development of Surfaces – 9 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/G4tKc2O1ZbU>

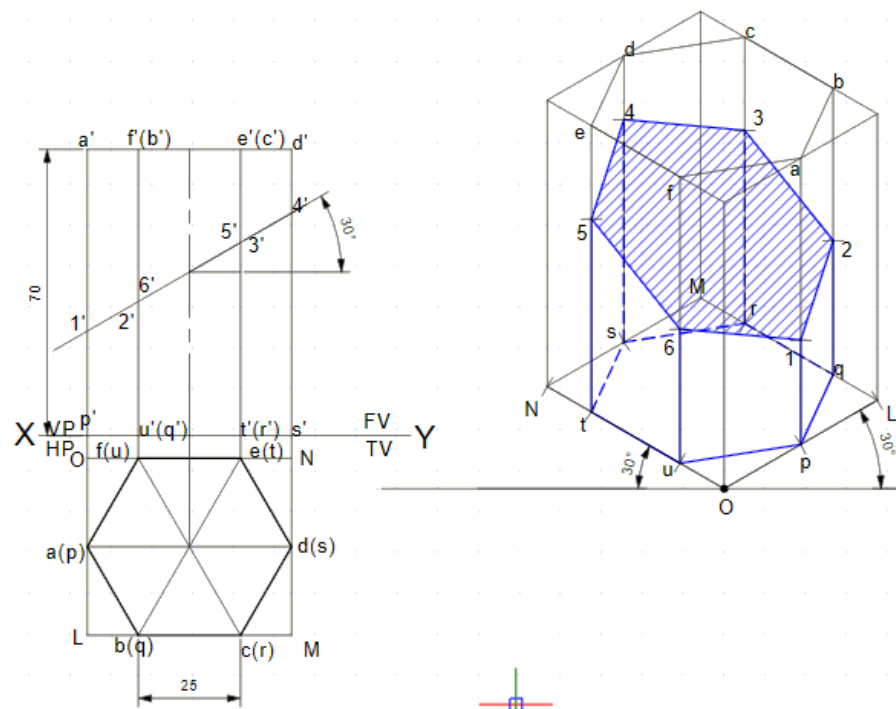


ISOMETRIC PROJECTION OF SOLIDS

1. A hexagonal prism, with a base of 25 mm and height 70 mm, rests on HP, and one of the edges of its base is parallel to VP. A section plane perpendicular to VP and inclined at 30° to HP cuts the axis of the prism at 30 mm from the top of the prism. Draw the isometric view of the truncated portion of the prism, clearly showing the cut surface.

Isometric Projections – 1 | Engineering Graphics Made Easy (CAD Method)

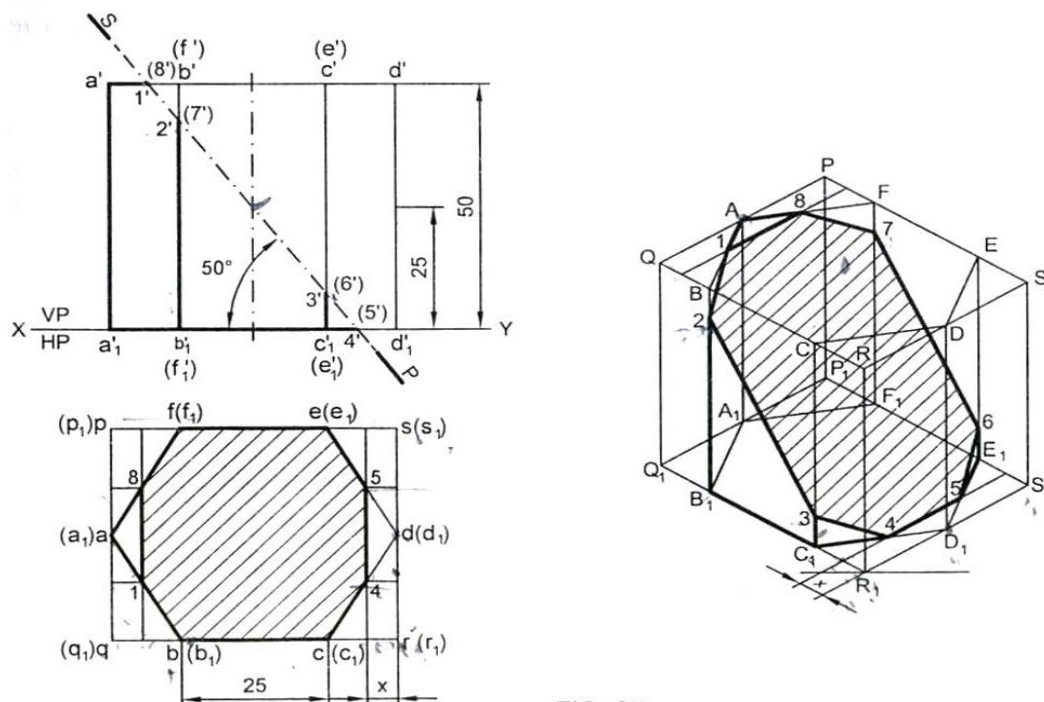
Link: <https://youtu.be/Wsg5-A0jbg>



2. A hexagonal prism, side of base 25 mm and height 50 mm rests on HP and one of the edges of its base is parallel to VP. A section plane perpendicular to VP and inclined at 50° to HP bisects the axis of the prism. Draw the isometric projection of the truncated prism. Showing the cut surface.

Isometric Projections – 2 | Engineering Graphics Made Easy (CAD Method)

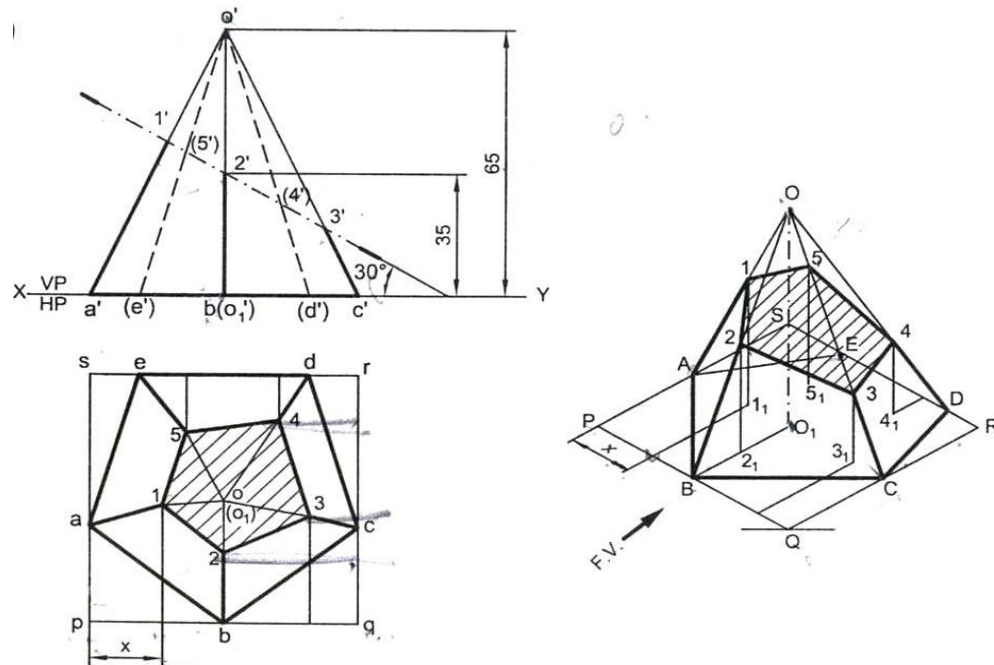
Link: https://youtu.be/_gHdl6bu4KQ



3. A pentagonal pyramid, 30 mm edge of base and 65 mm height, stands on HP such that an edge of the base is parallel to VP and nearer to it. A section plane perpendicular to VP and inclined at 30° to HP cuts the pyramid passing through a point on the axis at a height of 35 mm from the base. Draw the isometric projection of the truncated pyramid, showing the cut surface.

Isometric Projections – 3 | Engineering Graphics Made Easy (CAD Method)

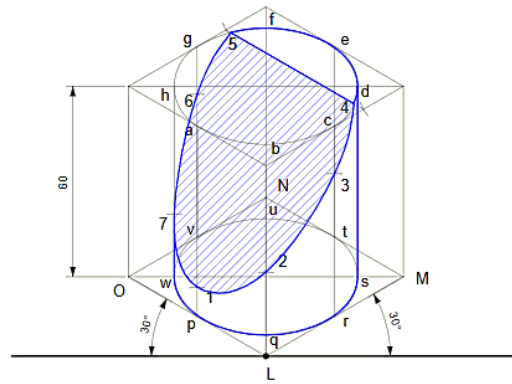
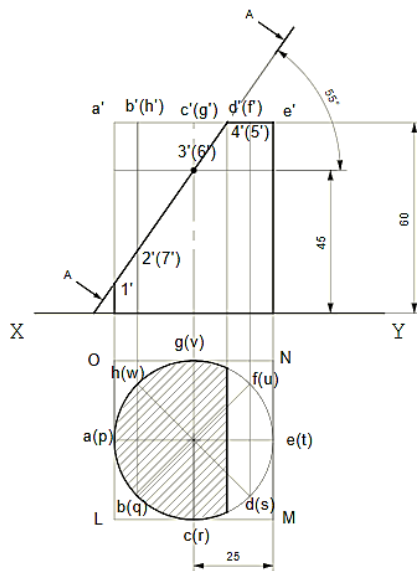
Link: <https://youtu.be/7NhRr6twmql>



4. A cylinder of 50 mm diameter and 60 mm height stands on HP. A section plane perpendicular to VP inclined at 55° to HP cuts the cylinder and passing through a point on the axis at a height of 45 mm above the base. Draw the isometric view of the truncated portion of the cylinder such that the cut surface is clearly visible to the observer.

Isometric Projections – 4 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/Bgcl0XW1NZQ>



ISOMETRIC SCALE
ALL DIMENSIONS ARE IN MM

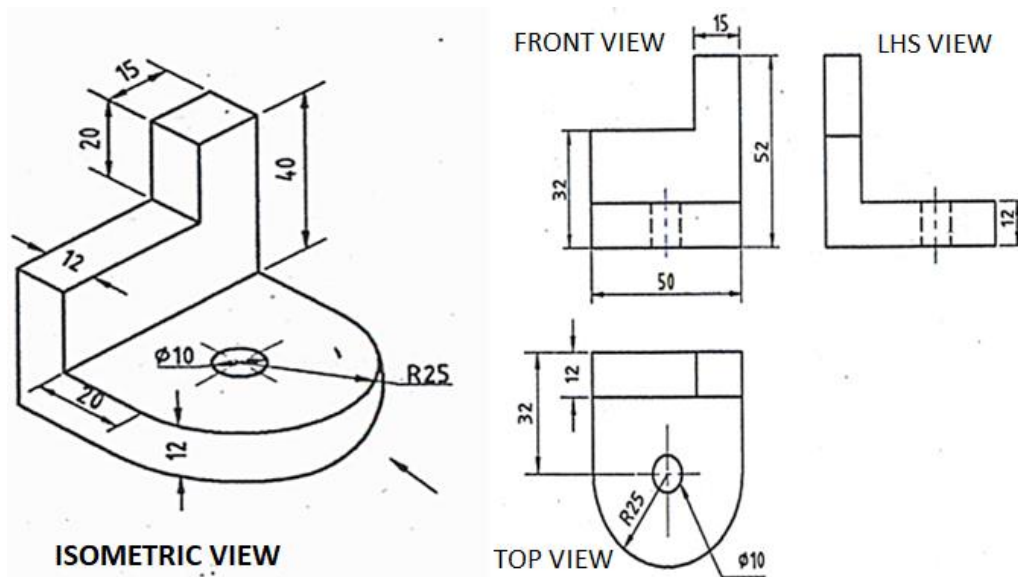
Unit – 5 – FREE HAND SKETCHING AND PERSPECTIVE PROJECTION

Question: Draw the Elevation (FV), Plan (TV) and Side views (More Visible side) of the Isometric drawings given below. Use appropriate scale if required. Take Elevation (Front View) as the arrow direction.

Question – 1

Free Hand Sketch – 1 | Engineering Graphics Made Easy (CAD Method)

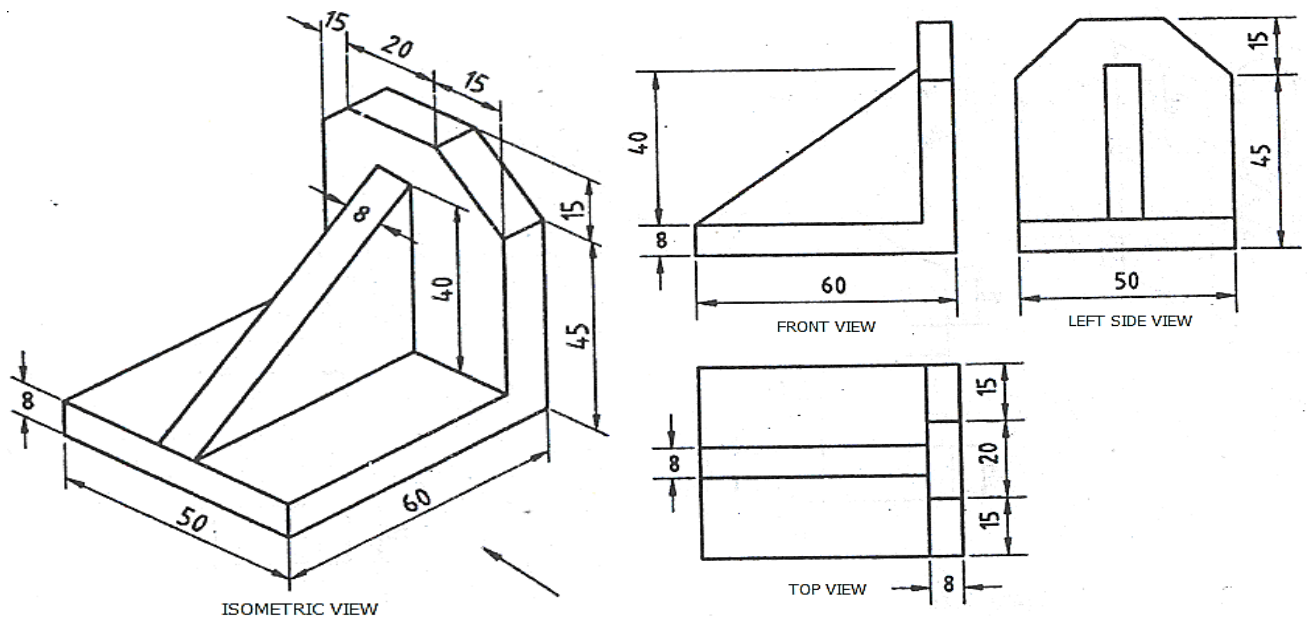
Link: <https://youtu.be/ywRUuU1N9VM>



Question – 2

Free Hand Sketch – 2 | Engineering Graphics Made Easy (CAD Method)

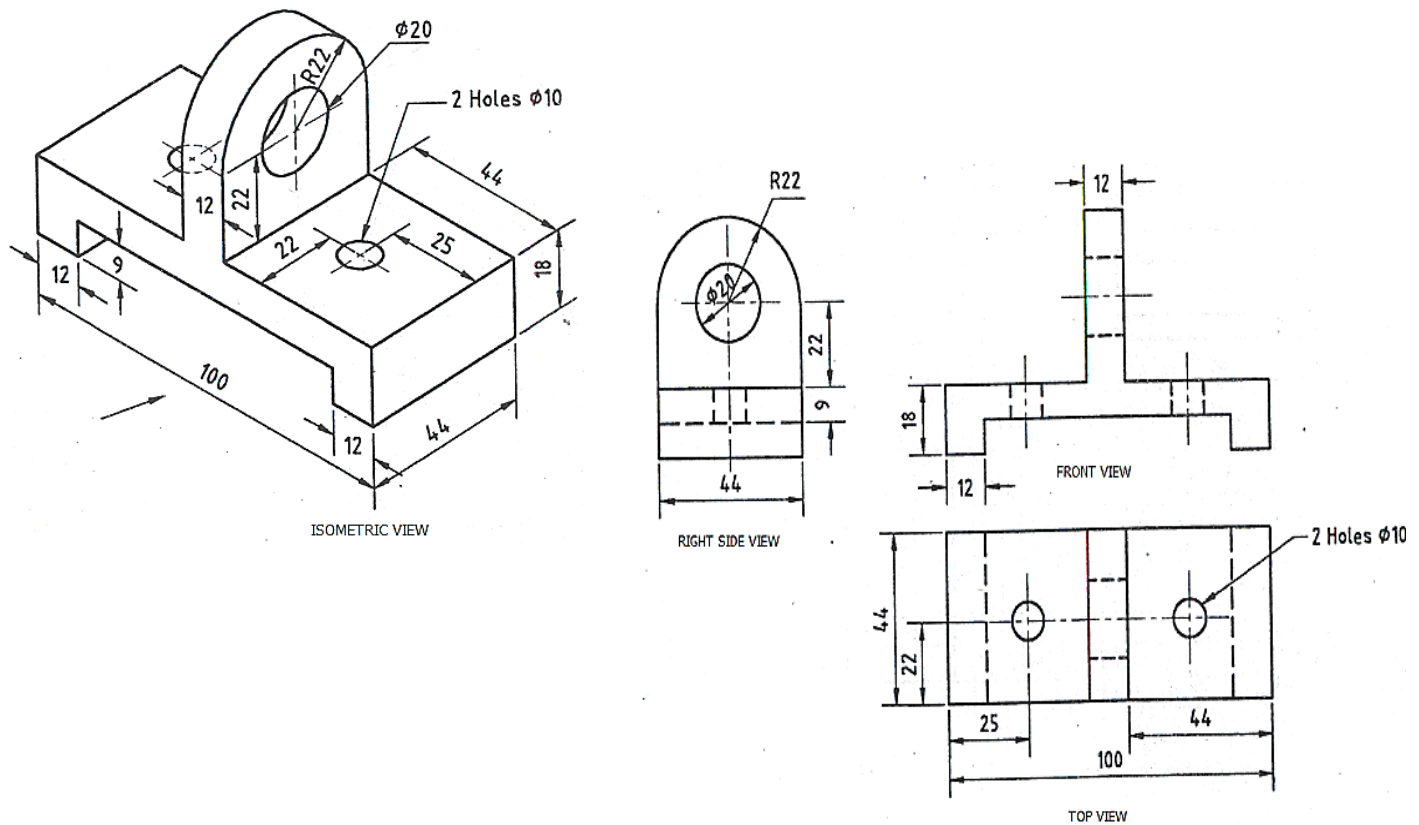
Link: <https://youtu.be/ho90Ddfgisc>



Question – 3

Free Hand Sketch – 3 | Engineering Graphics Made Easy (CAD Method)

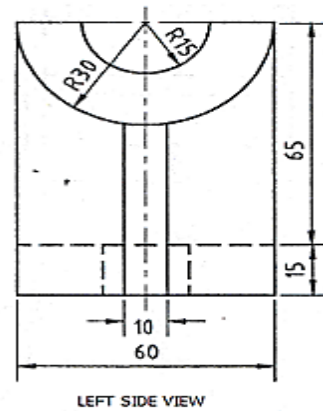
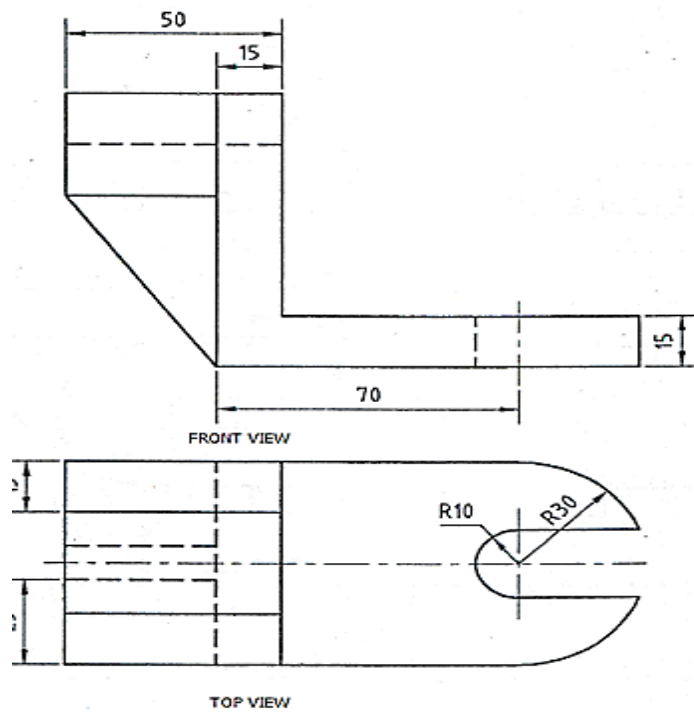
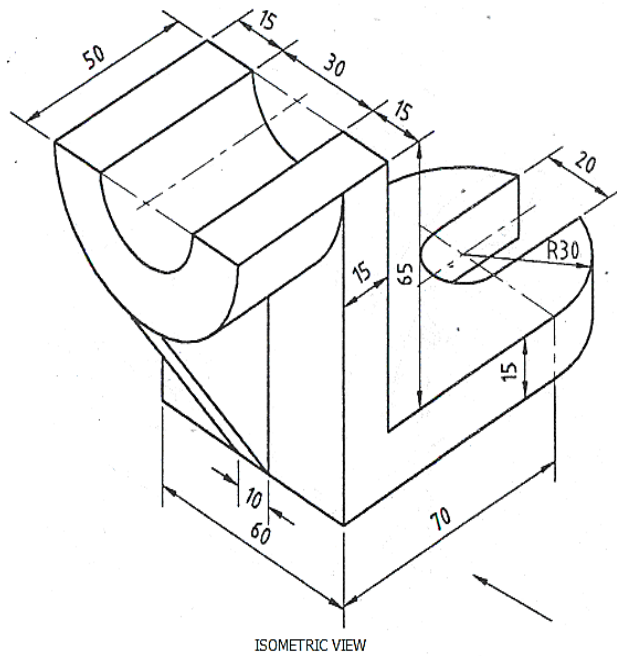
Link: <https://youtu.be/wSqAO3mtcJ0>



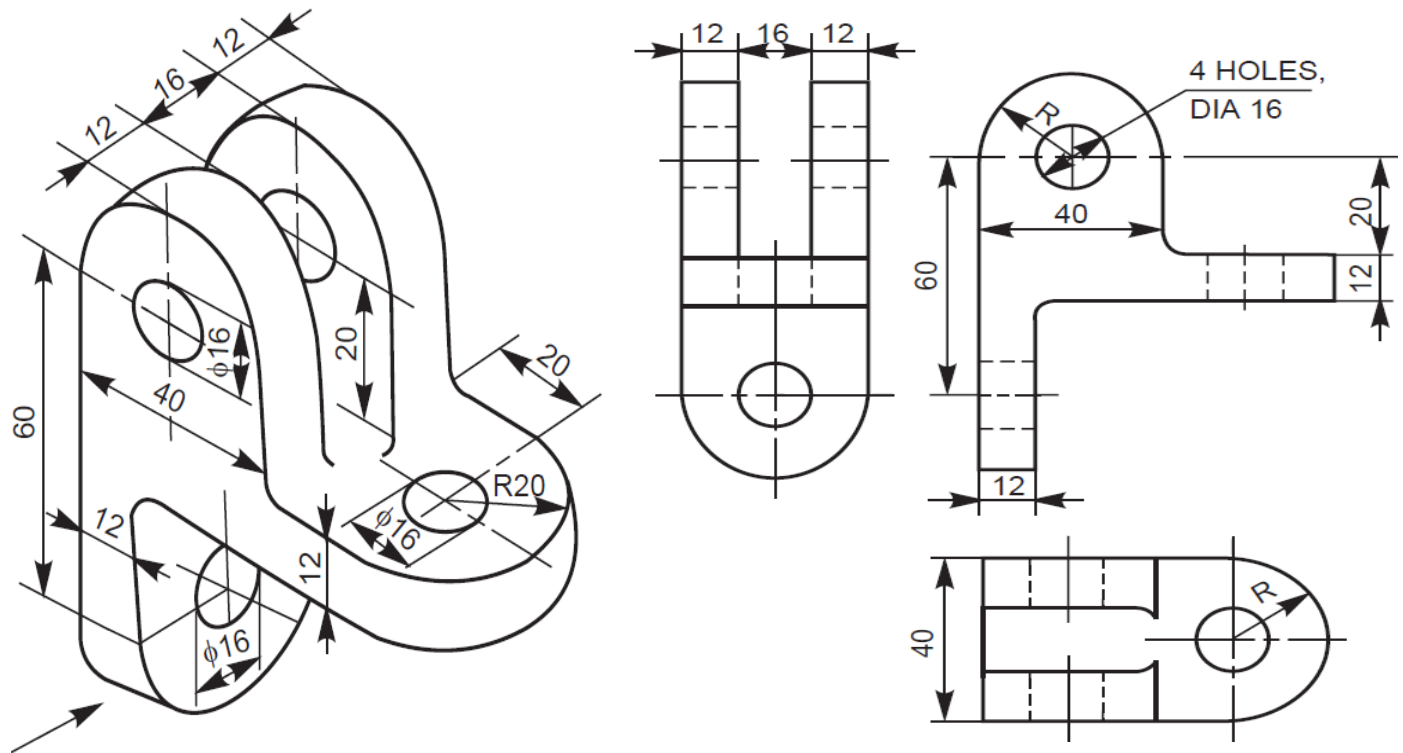
Question – 4

Free Hand Sketch – 4 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/8Q-3ynIM-Jk>



Problems for self-Practice:



PERSPECTIVE PROJECTION

1. A square pyramid of base edge 20 mm and altitude 40 mm rests on its base on the ground with a base edge parallel to the picture plane. The axis of the pyramid is 25 mm behind the PP and 25 mm to the right of the eye. The eye is 50 mm in front of the PP and 50 mm above the ground. Draw its perspective view. (Answer – 1)

Perspective Projection – 1 | Engineering Graphics Made Easy (CAD Method)

Link: https://youtu.be/g_WTEUk0eUs

2. A square prism, side of base 40 mm and a height 60 mm rests with its base on the ground such that one of its rectangular faces is parallel to and 10 mm behind the picture plane. The station point is 30 mm in front of PP, 80 mm above the ground plane, and lies in a central plane 45 mm to the right of the center of the prism. Draw the perspective projection of the square prism. (Answer – 2)

Perspective Projection – 2 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/JR8l77q6Ass>

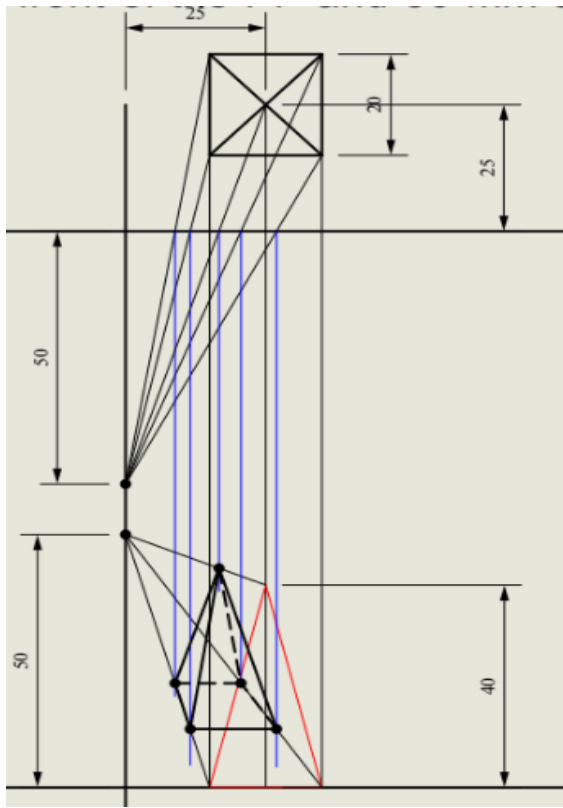


Figure – 1

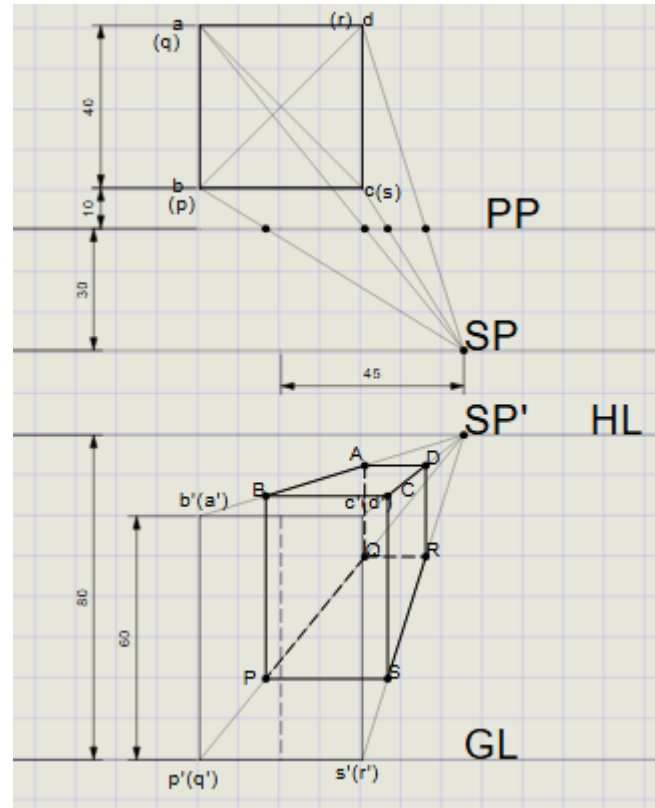


Figure – 2

3. A Hexagonal prism of base side 25 mm and height 50 mm is resting with its base on the GP such that one of its rectangular faces is inclined at 30° to the PP and the vertical edge nearer to PP is 15 mm behind it. The station point is 45 mm in front of the PP, 70 mm above the GP and lies in a central plane, which is 15 mm to the left of the vertical edge nearer to the PP. Draw the perspective view of the solid. (Answer – 3)

Perspective Projection – 3 | Engineering Graphics Made Easy (CAD Method)

Link: <https://youtu.be/IAV2LtqfA28>

4. A Hexagonal pyramid of base side 25 mm and axis length 50 mm is resting on the GP on its base with a side of base inclined at 30° to PP. The nearest corner to the PP is 20 mm behind it. The station point is 60 mm above the GP and 80 mm in front of PP and lies in a central plane, which is 50 mm to the left of the axis of the pyramid. (Answer – 4)

Perspective Projection – 4 | Engineering Graphics Made Easy (CAD Method)

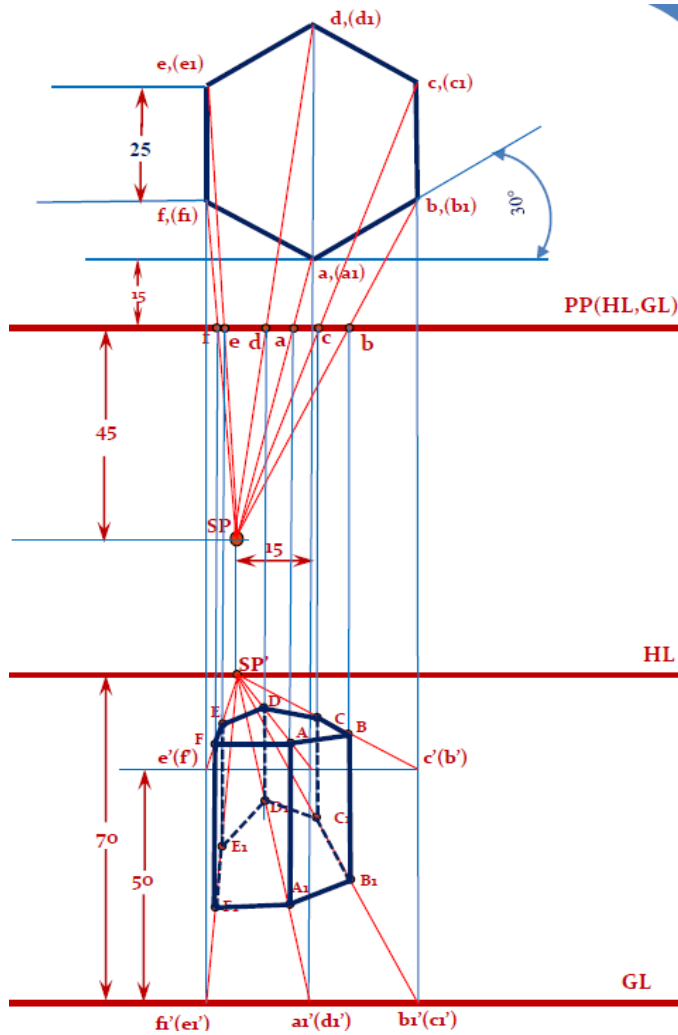
Link: https://youtu.be/VKF1izVT_3o

5. Draw the perspective view of a Pentagonal prism of base side 20 mm and height 40 mm when it rests on its base on the ground plane with one of the rectangular faces parallel to and 20 mm behind the picture plane. The station point is 45 mm

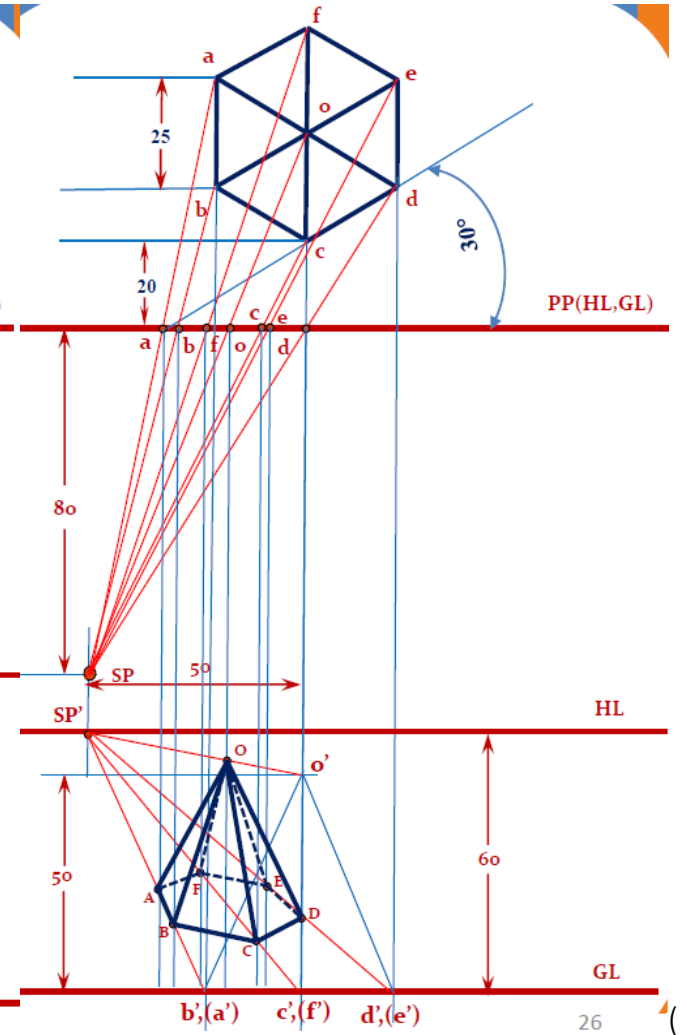
in front of the PP and 60 mm above the GP. The observer is 20 mm to the left of the axis. Use top view and the end view to draw the perspective by visual ray method. (Answer – 5)

Perspective Projection – 5 | Engineering Graphics Made Easy (CAD Method)

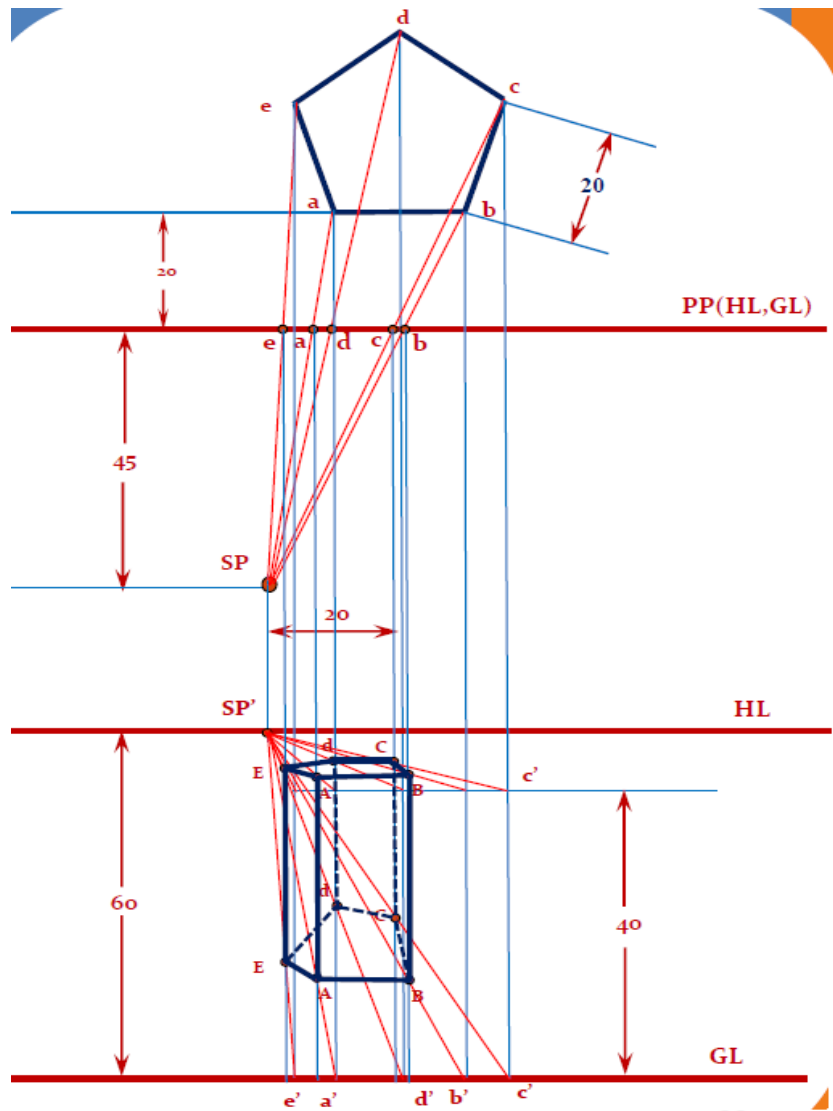
Link: <https://youtu.be/Mhf9k9o8lfY>



Answer – 3)



(Answer – 4)



(Answer – 5)